

Brachytherapy of Choroidal Melanomas Less Than 10 mm in Largest Basal Diameter

Comparison of 10-mm and 15-mm Ruthenium Plaques

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Purpose: To compare tumor control, vision, and complications between patients with a choroidal melanoma of <10 mm in largest basal diameter (LBD) irradiated with 10-mm or 15-mm ruthenium plaques.

Design: Retrospective, comparative case series.

Participants: One hundred sixty-four consecutive patients with a choroidal melanoma of <10 mm in LBD, 76 and 88 treated with the 10-mm and 15-mm plaque, respectively, from 1998–2014 in a national ocular oncology service.

Methods: Diagnosis was based on growth or high-risk characteristics. The apical dose was 100 to 120 Gy aiming to deliver ≥ 250 Gy to the sclera. Plaque positioning was modeled retrospectively. An increase of ≥ 0.3 mm in thickness and ≥ 0.5 mm in LBD indicated local recurrence. Outcomes were compared with cumulative incidence analysis and Cox regression. Median follow-up time for patients still alive was 8.4 years.

Main Outcome Measures: Recurrence rate, low vision, blindness, radiation maculopathy, and optic neuropathy.

Results: Melanomas treated with the 10-mm plaque were smaller (median thickness, 1.9 mm vs. 2.6 mm; LBD, 7.1 mm vs. 8.6 mm) and located closer to foveola (median, 2.0 mm vs. 2.8 mm) than those treated with the 15-mm plaque ($P < 0.001$). The 2 plaques provided a safety margin in 43% versus 40% eyes, provided no safety margin to guard foveola in 17% versus 33%, and did not entirely cover tumor mainly close to the disc in 32% versus 18% of eyes, respectively ($P = 0.052$). The incidence of a local recurrence was comparable (13% vs. 15% at 10 years; $P = 0.31$) and associated with plaque positioning (hazard ratio [HR], 2.81 for no safety margin; $P = 0.041$). At 5 years, the incidence of low vision was 14% versus 24%, and that of blindness was 3% versus 6%. Distance to the foveola was associated with loss of both levels of vision (HR, 0.65 per 1 mm vs. 0.68 per 1 mm; $P \leq 0.001$ vs. $P = 0.004$). The incidence of radiation maculopathy was comparable (19% vs. 18% at 5 years), whereas that of optic neuropathy tended to be higher with the 15-mm plaque (2% vs. 9%; $P = 0.054$).

Conclusions: The 10-mm ruthenium plaque contributes to better visual preservation, particularly with tumors close to fovea, without increase in local recurrence rate, and may therefore be preferable to the 15-mm plaque. *Ophthalmology* 2021;128:140-151 © 2020 by the American Academy of Ophthalmology



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Small choroidal melanomas are typically treated with eye-preserving treatments. Even small tumors are thought to be able to micrometastasize early; thus, enucleation is no longer a generally accepted alternative.¹ According to a recent study by the European Ocular Oncology Group, a critical size may be as small as 3 mm in largest basal diameter (LBD), and melanomas that metastasize when still small clinically are indistinguishable by their high-risk features from other melanomas of equal size.^{2–4} The primary aim when treating a small choroidal melanoma conservatively is to prevent local recurrences that are associated with a higher risk for metastases.⁵ An important secondary aim is to preserve as much useful vision as possible by minimizing side effects, especially radiation maculopathy and optic neuropathy.

One way of reducing the risk of radiation-related side effects is to decrease the size of the radioactive applicator. Compared with a 15-mm ruthenium plaque, which is used widely to irradiate small choroidal melanomas, the irradiated volume is smaller with a 10-mm plaque. A local tumor control rate of 91% at 5 years and cumulative incidence rate of avoiding low vision and blindness of 83% and 97%, respectively, in our previous noncomparative case series of 45 patients provided preliminary support for the use of the 10-mm ruthenium plaque in treating choroidal melanomas of less than 10 mm in LBD.⁶ Our present comparative series explored whether the 10-mm plaque has an advantage over the 15-mm plaque in treating small choroidal melanomas of this size, adjusting for differences in baseline characteristics by multivariate regression analysis. We compared local

tumor control, vision outcome, and side effects between 76 and 88 consecutive patients who were managed with the 10-mm and 15-mm ruthenium plaques, respectively.

Methods

Aims of the Study

The primary aim was to compare local tumor control, best-corrected visual acuity (BCVA), and complications between eyes irradiated with the 10-mm and 15-mm ruthenium plaque in a population-based consecutive series of patients with a choroidal melanoma of less than 10 mm in LBD. The secondary aim was to compare mortality rates between the 2 groups.

Inclusion Criteria

Eligible to our study were patients diagnosed with a choroidal melanoma of less than 10 mm in LBD between October 1998 and December 2014 who were scheduled to receive a 10-mm (code: CCX) or 15-mm (code: CCA) ruthenium plaque (Eckert & Ziegler BEBIG GmbH, Berlin, Germany). The LBD was based on indirect ophthalmoscopy and B-scan echography (I3 System-ABD; Innovative Imaging, Sacramento, CA). The 10-mm ruthenium plaque came into use in 1998 in the Ocular Oncology Service, Department of Ophthalmology, Helsinki University Hospital, Finland, a national referral center for uveal melanomas, and gradually became the plaque of choice for tumors of this size, especially those close to the foveola. In the early years, we more often chose the 15-mm plaque for tumors close to 10 mm in LBD, and the 15-mm plaque was additionally used for 13 patients to expedite treatment when the 10-mm plaque was in use for another patient. The study followed the tenets of the Declaration of Helsinki and was approved by the institutional review board of the Hospital Region of Helsinki and Uusimaa. This was a retrospective study, and the requirement for informed consent was waived by the Review Board.

We reviewed the database and identified 234 patients with a choroidal melanoma of less than 10 mm in LBD, of whom 76 had been scheduled for the 10-mm and 88 for the 15-mm plaque ruthenium radiotherapy. One patient received the 10-mm plaque as secondary treatment after regrowth of a tumor treated 20 years earlier with 3 sessions of argon laser photocoagulation, and 1 patient received the 15-mm plaque 7 months after photodynamic therapy. All tumors irradiated with the 10-mm plaque and 16 of the 88 treated with the 15-mm plaque were T1aN0M0, stage I, and the remaining tumors were T2aN0M0, stage IIA, by the American Joint Committee on Cancer's tumor, node, metastasis system, eighth edition.⁷ By Collaborative Ocular Melanoma Study criteria, 65 (86%) and 60 (68%) tumors, respectively, would have been classified as small.⁸ An additional 70 tumors of the same size were scheduled for other treatments (Fig S1 and Table S1, available at www.aaojournal.org, which show types of various plaques used and tumor dimensions by plaque type).

Clinical Evaluation

We measured the BCVA using a test-type projector (Rodavist 2 and 524; Rodenstock GmbH, Ottobrunn, Germany) in even decimals from 1.0 (20/20) to 0.1 (20/200) plus 0.05 (20/400). Best-corrected visual acuity worse than 20/400 was recorded as counting fingers at 2 m, 1 m, and 0.5 m; hand movements; light perception; and no light perception. We used biomicroscopy, indirect ophthalmoscopy, and fundus photography to localize the tumor margins, estimate the distance from the posterior tumor margin to the optic disc and foveola, and evaluate the growth pattern, presence of subretinal fluid, and presence of orange pigment.

We treated the patients if they had a presumed small choroidal melanoma because it had been observed to grow (37%) or had high-risk characteristics predictive of growth (Table 2): thickness of more than 2 mm in 93 eyes (57%), subretinal fluid in 103 eyes (63%), symptoms in 79 eyes (48%), orange pigment in 92 eyes (56%), and tumor margin within 2 disc diameters of the optic disc in 81 eyes (49%), of which 4 tumors (2%) touched the disc margin (Table 2). Symptoms included decreased vision, a visual field defect, flashes, floaters, and metamorphopsia. Other risk characteristics were ultrasound hollowness in 113 eyes (69%), absence of a surrounding halo in 161 eyes (98%), and absence of drusen in 113 eyes (69%). The range of observed growth was 0.1 to 2.3 mm (median, 0.5 mm) in thickness and 0.1 to 6.0 mm (median, 1.0 mm) in LBD. Twelve tumors (7%) were treated without any risk characteristics because of growth. Two patients (2%) were treated without risk factors or growth. The first one had a tumor that had not been seen during regular visits for glaucoma and was hollow by ultrasound. The second one had a ruptured Bruch's membrane.

Brachytherapy

The 10-mm and 15-mm plaques have a physical diameter of 11.6 and 15.3 mm, respectively, and a radius of curvature of 12 mm. The corresponding diameters of the irradiation windows are 10.0 mm and 13.8 mm before reshaping the applicator to spherical contour. The window is surrounded by a 0.75-mm wide inactive border. In the spherical form, the cord diameter of the radiation window is 9.4 and 12.9 mm, respectively, surrounded by a penumbra of scatter radiation that facilitates eccentric placement to limit irradiating adjacent tissues.^{9,10} The prescription point was thickness plus 1 mm for the sclera. The apical dose was 100 to 120 Gy to tumor apex aiming to deliver a minimum scleral dose of 250 Gy. Treatment times and doses were calculated from the central axis depth dose curve (Table 3), and calculations were based on the Bebig Plaque Simulator data version 5.3.7. The inferior oblique muscle was detached in 26 eyes (34%) with the 10-mm plaque and 39 eyes (44%) with the 15-mm plaque when the tumor was located in the macular area. Three experienced ocular oncologists (T.T.K. and V.R.) performed the treatments. The surgical technique has been previously described.⁶

Assessment of Local Tumor Control, Metastasis, Vision, and Side Effects

We followed up patients prospectively at 3, 6, and 12 months and every 6 months until 3 to 4 years. Thereafter, 8 patients were seen annually in a central hospital, 19 patients were seen at other universities, and the remaining patients were seen in our service. We used biomicroscopy, indirect ophthalmoscopy, fundus photography, and B-scan echography to assess tumor control. We coded recurrences as marginal, vertical, diffuse (marginal plus vertical), or extrascleral.¹¹ An unequivocal change of at least 0.3 mm and 0.5 mm in thickness and LBD indicated a vertical and marginal recurrence, respectively.

We retrospectively modeled the location of the plaque to facilitate determining the reasons for local recurrence. We used the radiation and diathermy scars, if visible, on posttreatment fundus photographs together with the surgical notes to draw digitally on a color fundus photograph a circle corresponding to the radiation window, presuming the average diameter of the optic disc to be 1.5 mm. We coded the location of the radiation window relative to tumor margins as covering the entire tumor with a safety margin, being parallel, or not covering the entire tumor.⁶

We performed liver function tests and upper abdominal ultrasonography at the time of diagnosis and annually thereafter and a chest radiogram at diagnosis. Suspicious lesions were

Table 2. Risk Characteristics for Growth in 164 Small Choroidal Melanomas of Less than 10 mm in Largest Basal Diameter Managed with 10-mm and 15-mm Ruthenium Plaques

Risk Characteristics	Plaque Diameter			Observed Growth		
	10 mm (n = 76)	15 mm (n = 88)	P Value	10 mm (n = 29)	15 mm (n = 32)	P Value
Thickness >2 mm	30 (39)	63 (72)	<0.001*	6 (21)	19 (59)	0.004*
Subretinal fluid	48 (63)	55 (63)	0.99*	8 (28)	11 (34)	0.81*
Symptoms	40 (53)	39 (44)	0.35*	8 (28)	5 (16)	0.35*
Orange pigment	47 (62)	45 (51)	0.21*	13 (45)	11 (34)	0.44*
Margin						
Touching disc	3 (4)	1 (1)	0.34*	1 (1)	0 (0)	0.48*
Within 2 DD from disc	40 (53)	41 (47)	0.53*	8 (28)	14 (44)	0.19*
No. of characteristics						
0	9 (12)	5 (6)	0.87 [†]	8 (28)	4 (13)	0.12 [†]
1	9 (12)	10 (11)	7 (24)	7 (22)		
2	16 (21)	17 (19)	8 (28)	11 (34)		
3	14 (18)	30 (34)	4 (14)	9 (28)		
4	19 (25)	21 (24)	2 (7)	1 (1)		
5	9 (12)	5 (6)	0 (0)	0 (0)		

DD = disc diameters.

*Fisher exact test.

[†]Nonparametric test for trend.

verified and staged with computed tomography, magnetic resonance imaging, or both. A sample for histopathologic confirmation was obtained with a fine-needle or core-needle biopsy. Causes of death were verified from death certificates and patient charts.

We assessed the visual outcome after the conventions of the Collaborative Ocular Melanoma Study.¹² Measurements worse than 20/1600, such as counting fingers or hand movements, were set to 20/1600. Enucleated eyes and eyes with no light perception were counted as having 20/2500 vision. We analyzed low vision (BCVA, <20/65) and blindness (BCVA, <20/400) according to the World Health Organization¹³ standard definitions. Loss of BCVA was recorded when confirmed at the next review. If a measurement had not been obtained at every review, we considered the BCVA to be the mean of the previous and the next visit. Otherwise, we coded the data as missing. We assessed side effects as outlined in a previous study from our service.¹⁴ Maculopathy was categorized as radiation maculopathy or other maculopathy (recurrent tumor, radiation scar, epiretinal membrane, tumor-related lipid exudation, and exudative retinal detachment).

Statistical Methods

We collected follow-up data prospectively up to the date of analysis on December 31, 2017, and analyzed them with Stata

(release 15; Stata Corp, College Station, TX). We analyzed regression in tumor thickness at 6-month intervals. We used cumulative incidence analysis that takes competing risks into account to provide estimates that would be most useful in clinical counseling in comparing outcomes and compared groups with the Gray k-sample test. We used the Fisher exact test in comparing dichotomous variables, a nonparametric test for trend to explore differences between 2 independent groups for ordinal variables, and the Kruskal-Wallis test to explore differences between 2 or more independent groups for continuous variables. All tests were 2-tailed, and $P < 0.05$ was considered statistically significant.

We calculated mean BCVA after transformation to a logarithmic scale, stratified by factors associated with vision outcomes at 6-month intervals. We used multivariate Cox regression analysis to estimate the hazard ratio (HR), adjusting the effect of plaque type for previously identified explanatory and confounding variables for local recurrence and losing BCVA of 20/65 and 20/400. Variables significant at the univariate level were entered into bivariate and trivariate models. The assumption of proportional hazards was tested by the method of Therneau and Grambsch¹⁵ both globally and for individual covariates in the model. Independent variables were allowed in the model if the P value was less than 0.10. Models were compared using the likelihood ratio test.¹⁶ The

Table 3. Delivered Dose, Dose Rate, and Treatment Time for 164 Small Choroidal Melanomas of Less than 10 mm in Largest Basal Diameter Irradiated with the 10-mm and 15-mm Ruthenium Plaque*

Plaque	Apex		Sclera		Treatment Time (hrs)
	Dose (Gy)	Dose rate (cGy/hr)	Dose (Gy)	Dose Rate (cGy/hr)	
10 mm	117 (80–197)	11.5 (3.1–42.4)	346 (201–824)	36.0 (8.9–94.4)	98 (28–334)
15 mm	105 (79–183)	10.6 (3.4–40.0)	340 (169–1015)	38.0 (12.6–92.1)	102 (31–269)
P value [†]	0.060	0.66	0.65	0.89	0.89

Data are median (range) unless otherwise indicated.

*The standard prescription dose was either 100 or 120 Gy to the tumor apex depending on the thickness of the tumor, aiming to deliver a minimum dose of 250 Gy to the sclera.

[†]Kruskal-Wallis test.

best-fitting models are reported as final ones. We performed sensitivity analyses of the final Cox regression models for the 3 main outcomes—local tumor recurrence and development of low vision and blindness—by restricting LBD to less than 9 mm to balance the distribution of tumor diameter between the 2 study groups to further guard against the possibility that more frequent irradiation of tumors with LBD between 9.0 and 9.9 mm with the 15-mm plaque might have biased our estimates.

To guide the reader, we provide post hoc estimates of HR likely to be detected with 0.80 power for our 3 main outcomes, calculated for log-rank test using the Schoenfeld method. We set the significance level at 0.05, and we set the sample sizes and 10-year survival proportions to match those observed in each analysis.

Results

The median age at diagnosis (65 vs. 63 years; $P = 0.59$, Kruskal-Wallis test) and BCVA ($P = 0.55$, nonparametric test for trend) were comparable between the 2 groups (Table S4, available at www.aaojournal.org, which shows characteristics of patients and tumors). At the time of diagnosis, 6 eyes (8%) and 6 eyes (7%) showed low vision, and 1 eye (1%) and 3 eyes (3%) were blind by World Health Organization criteria¹³ in patients treated with the 10-mm and 15-mm plaque, respectively.

The melanomas treated with the 10-mm plaque were thinner (median, 1.9 mm vs. 2.6 mm; $P < 0.001$, Kruskal-Wallis test) and had a smaller LBD (median, 7.1 mm vs. 8.6 mm; $P < 0.001$) than those treated with the 15-mm plaque (Fig 2A, B). Distance from the posterior tumor margin to the foveola was shorter in eyes treated with the 10-mm plaque (median, 2.0 mm vs. 2.8 mm; $P < 0.001$; Fig 2C), so that the posterior tumor margin involved the foveola in 20 eyes (26%) treated with the 10-mm plaque and 17 eyes (19%) treated with the 15-mm plaque, and extended to within 2 disc diameters of it in 50 eyes (66%) treated with the 10-mm plaque and 50 eyes (57%) treated with the 15-mm plaque. The distance from the tumor margin to the optic disc was comparable (median for both, 3.0 mm; Fig 2D), as was the extent of subretinal fluid ($P = 0.59$, nonparametric test for trend).

Plaque Positioning

According to the surgical charts, the physical 10-mm and 15-mm plaque covered the tumor with a safety margin in 25 eyes (33%) and 50 eyes (57%), was eccentric in 28 eyes (37%) and 28 eyes (32%), and did not entirely cover the tumor in 22 eyes (29%) and 10 eyes (11%), respectively ($P = 0.002$, nonparametric test for trend). Retrospective modeling of coverage by the radiation window differed to a lesser extent ($P = 0.052$). The corresponding numbers were 33 eyes (43%) and 35 eyes (40%), 13 eyes (17%) and 29 eyes (33%), and 24 eyes (32%) and 16 eyes (18%). Modeling of plaque location was unsuccessful in 6 eyes (8%) and 7 eyes (8%) treated with the 10-mm and 15-mm plaque, respectively. No adverse events occurred during the surgical procedure.

Tumor Regression and Local Recurrence

The follow-up time for patients who were still alive was comparable (median, 8.1 years [range, 3.0–18.8 years] and 8.9 years [range 1.1–17.9 years], respectively, for eyes treated with the 10-mm and 15-mm plaque; $P = 0.45$, Kruskal-Wallis test). By 3 years, tumor thickness had reduced equally in the 2 groups (median, 42% and 35% in the 10-mm and 15-mm plaque groups, respectively; Fig 3A). A local recurrence was detected in 9 and 16 patients after a median follow-up of 1.5 years (range, 0.5–6.5 years) and 3.0 years (range, 0.3–14.0), respectively (Table S5,

available at www.aaojournal.org, which shows details of all recurrences). The cumulative incidence of a local recurrence developing (Fig 3B) was comparable: 9% (95% confidence interval [CI], 4%–17%) and 13% (95% CI, 7%–20%) at 5 years and 13% (95% CI, 6%–21%) and 15% (95% CI, 8%–23%) at 10 years, respectively ($P = 0.31$, Gray k-sample test). With 164 patients, we had 0.80 power post hoc to detect an HR of 2.4 as statistically significant.

By univariate Cox regression, a local recurrence was associated with plaque positioning (HR, 2.81 for plaque not covering the tumor with a safety margin; $P = 0.041$; Table 6). In bivariate models, positioning the plaque predicted a local tumor recurrence (HR, 2.72; $P = 0.048$). In univariate models stratified by plaque size, the HRs (HR, 2.31 and 2.99 for the 10-mm plaque and 15-mm plaque, respectively) remained comparable (Tables 6 and S7, available at www.aaojournal.org, which shows alternative models). The models suggest that the combined effect derived predominantly from treatment with the 15-mm plaque. Sensitivity analysis, restricting LBD to less than 9 mm, showed that the HR for plaque size remained consistent with that in the full sample (HR, 1.91 vs. 2.03, respectively), whereas that for positioning the plaque became less influential (HR, 1.98 vs. 2.72, respectively; Fig S4 and Table S8, available at www.aaojournal.org, which show the size distribution between the 2 groups and comparison of HR in final models, respectively).

The recurrences were treated with a secondary iodine plaque, except for 1 that was retreated with a ruthenium plaque and 2 that received transpupillary thermotherapy (Table S5, which shows details of all recurrences). One of 9 patients treated with the 10-mm plaque and 4 of the 16 patients (25%) treated with the 15-mm plaque demonstrated additional recurrences managed mostly with transpupillary thermotherapy (Table S5).

Metastasis and Survival

Of patients treated with the 10-mm and 15-mm plaque, 16 and 28 died during follow-up, 3 and 7 of them of metastatic choroidal melanoma, respectively. One patient was alive with metastases after a local recurrence (Table S5, case 4). Nine of the 11 metastases were verified histologically and 2 were diagnosed with imaging (a solitary brain tumor classified as a possible metastasis, and multiple hepatic lesions, a fine-needle aspiration biopsy of which failed). All 4 patients who demonstrated metastases after the 10-mm plaque and 4 of the 7 patients treated with the 15-mm plaque had experienced a local recurrence a median of 1.8 years (range, 0–5.7 years) and 1.5 years (range, 0–3.5 years) before metastases, respectively. Conversely, 4 of 9 patients (44%) and 4 of 16 patients (25%) with a local recurrence later demonstrated metastases, respectively.

The cumulative incidence of metastases developing was 4% (95% CI, 1%–10%) at 5 and 10 years after treatment with the 10-mm plaque and 6% (95% CI, 2%–13%) at 5 years and 10% (95% CI, 4%–19%) at 10 years after treatment with the 15-mm plaque ($P = 0.50$, Gray k-sample test). The cumulative incidence estimate of all-cause mortality was 10% (95% CI, 4%–18%) and 9% (95% CI, 4%–17%) at 5 years and 24% (95% CI, 13%–37%) and 29% (95% CI, 19%–40%) at 10 years, respectively (Fig 3C; $P = 0.24$). The cumulative incidence estimate of melanoma-related mortality was 3% (95% CI, 1%–8%) at 5 and 10 years and 2% (95% CI, 0.5%–7%) at 5 years and 8% (95% CI, 3%–15%) at 10 years, respectively (Fig 3D). These percentages were comparable and far exceeded by deaths resulting from other causes. The time from treatment of the primary tumor to metastases (median, 4.6 vs. 4.5 years; $P = 0.85$, Kruskal-Wallis test) and survival after metastases (median, 3.6 vs. 6.1 years; $P = 0.73$) were also comparable between the 2 groups.

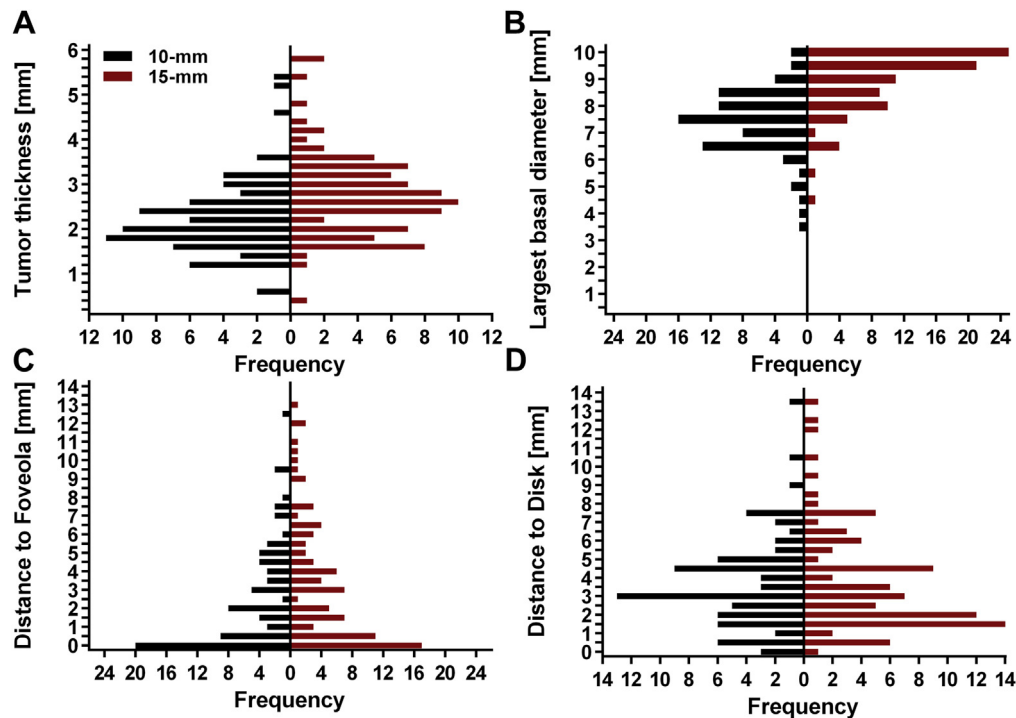


Figure 2. Graphs showing frequency distribution of (A) tumor thickness, (B) largest basal diameter, (C) distance to foveola, and (D) distance to optic disc margin of choroidal melanomas less than 10 mm in largest basal diameter, of which 76 were irradiated with the 10-mm (black) and 88 were irradiated with the 15-mm (red) ruthenium plaque.

Preservation of Vision

The median BCVA at 5 years was 20/32 and 20/30 and at 10 years 20/40 and 20/40 in eyes treated with the 10-mm and 15-mm plaque, respectively. Five years after diagnosis, the BCVA was at least 2 lines better in 5 eyes (8%; 95% CI, 3%–18%) and 11 eyes (16%; 95% CI, 8%–27%), within 1 line of pretreatment BCVA in 22 eyes (37%; 95% CI, 25%–50%) and 18 eyes (26%; 95% CI, 17%–39%), and at least 2 lines worse in 33 eyes (55%; 95% CI, 42%–68%) and 39 eyes (57%; 95% CI, 45%–69%; $P = 0.88$, nonparametric test for trend), respectively.

Tumor thickness of more than 3.0 mm was associated with loss of BCVA, more notably after the 10-mm plaque (Fig 5A, B), whereas increasing LBD did not predict loss of vision (Fig 5C, D). Location of the tumor margin less than 1.5 mm from the foveola (Fig 5E, F) was associated with mild loss of BCVA, especially after treatment with the 15-mm plaque, as was a distance of less than 4.5 mm from the optic disc with the 15-mm plaque (Fig 5G, H).

The cumulative incidence of losing 20/65 vision in 70 and 82 eyes that had at least this BCVA at diagnosis was 14% (95% CI, 7%–24%) and 24% (95% CI, 15%–34%; Fig 3E), and that of losing 20/400 vision in 75 and 85 eyes was 3% (95% CI, 0.6%–10%) and 6% (95% CI, 2%–13%; Fig 3F) at 5 years with the 10-mm and 15-mm plaque, respectively. The corresponding incidences at 10 years after diagnosis were 21% (95% CI, 11%–33%) and 30% (95% CI, 20%–41%) for low vision and 14% (95% CI, 6%–25%) and 13% (95% CI, 6%–23%) for blindness, respectively. With 152 and 160 patients, we had 0.80 power post hoc to detect an HR of 2.0 and 2.3 as significant regarding low vision and blindness developing, respectively.

By univariate Cox regression, low vision and blindness were associated with distance to the foveola (HR, 0.65 and 0.68 for each 1-mm increase for the 10-mm and 15-mm plaque groups;

$P < 0.001$ and $P = 0.004$, respectively; Table 9), baseline BCVA (HR, 0.16 and 0.3 for each 0.1-logarithm of the minimum angle of resolution decrease; $P < 0.001$ and $P = 0.043$, respectively), and tumor thickness (HR, 1.31 and 1.53 for each 1-mm increase; $P = 0.077$ and $P = 0.025$, respectively). Of bivariate models that included plaque size, the one with distance to the foveola best predicted low vision (HR, 0.64 for each 1-mm increase in distance; $P < 0.001$; Table 9). A trivariate model that additionally included BCVA at baseline or tumor thickness did not provide a significantly better fit ($P = 0.23$ and $P = 0.094$, respectively, likelihood ratio test). Distance to the foveola was also independently associated with blindness developing in a bivariate model (HR, 0.68 for each 1-mm decrease in distance; $P = 0.004$), and a trivariate model that additionally included tumor thickness fitted best with the data ($P = 0.010$, likelihood ratio test).

Sensitivity analysis, restricting LBD to less than 9 mm, showed that the HR for plaque size remained close to that in the full sample (HR, 1.56 vs. 1.79, respectively), as did that for tumor distance to the foveola (HR, 0.60 vs 0.64 for a 1-mm change) for low vision developing (Fig S4 and Table S8, available at www.aaojournal.org, which show the size distribution between 2 groups and a comparison of HR in the final models, respectively). Because plaque size was not nearly a significant predictor of blindness developing ($P = 0.69$), its HR estimate is unstable in the sensitivity analysis and full sample. The effect of tumor distance to the foveola is estimated to be slightly stronger and the effect of tumor thickness slightly less (Table S8).

Enucleation and Cosmetic Outcome

No secondary enucleation was performed after primary treatment. One patient demonstrated extraocular extension after treatment for multiple local recurrences (Table S5, case 4), and 1 eye was blind and painful as a result of secondary glaucoma after vitrectomy for

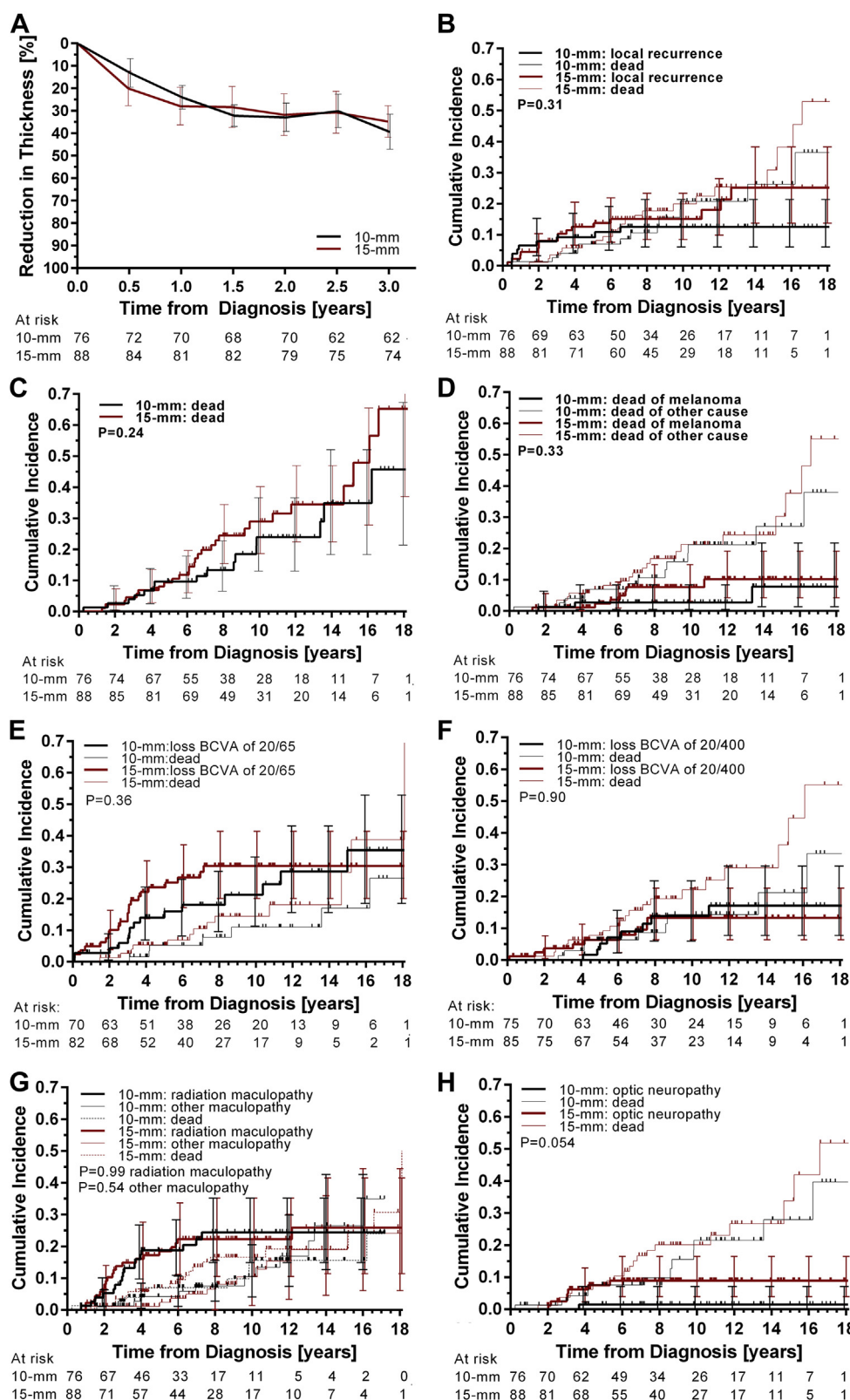


Figure 3. Graphs showing (A) percentage reduction in thickness for 76 and 88 choroidal melanomas less than 10 mm in largest basal diameter irradiated with the 10-mm (black) and 15-mm (red) ruthenium plaque, respectively, (B) cumulative incidence of local recurrence, (C) death, (D) melanoma-related death, (E) low vision, (F) blindness by World Health Organization definitions, (G) maculopathy, and (H) optic neuropathy. The main competing risks also are plotted: (C, B, F–H) death of any cause, (D) death of causes other than melanoma, and (G) other maculopathy. Crossbars show 95% confidence intervals, and ticks show censored observations. Numbers are patients at risk. BCVA = best corrected visual acuity.

Table 6. Cox Proportional Hazards Regression of Time to Local Recurrence of 164 Patients with a Small Choroidal Melanoma Irradiated with the 10-mm and 15-mm Ruthenium Plaque

Variable	Coefficient (Standard Error)	Wald Chi-Square Test	P Value	Hazard Ratio (95% Confidence Interval)
Univariate analysis				
Plaque*	0.435 (0.417)	1.09	0.30	1.55 (0.68–3.50)
Thickness [†]	0.255 (0.191)	1.78	0.18	1.29 (0.89–1.88)
LBD [‡]	0.128 (0.153)	0.70	0.40	1.14 (0.84–1.53)
Dfov [‡]	−0.123 (0.081)	2.31	0.13	0.88 (0.75–1.04)
Ddisc [‡]	−0.111 (0.091)	1.49	0.22	0.90 (0.75–1.07)
Dose to apex [‡]	−0.114 (0.115)	0.99	0.32	0.89 (0.71–1.12)
Dose to sclera [‡]	0.034 (0.125)	0.08	0.78	1.04 (0.81–1.32)
Eccentricity [§]				
Parallel	1.223 (0.541)	5.11	0.024	3.40 (1.18–9.80)
Not covered	0.797 (0.586)	1.85	0.17	2.22 (0.70–7.00)
Positioning	1.035 (0.506)	4.16	0.041	2.81 (1.04–7.59)
Bivariate analysis [¶]				
Plaque*	0.710 (0.454)	2.44	0.12	2.03 (0.83–4.96)
Positioning	1.000 (0.507)	3.90	0.048	2.72 (1.01–7.35)

Ddisc = distance to optic disc; Dfov = distance to foveola; LBD = largest basal diameter.

*Categories: 0 = 10-mm plaque; 1 = 15-mm plaque.

[†]Continuous variable (mm).

[‡]Continuous variable (10 Gy for apex, 100 Gy for sclera).

[§]Remodeling; categories: 0 = covered; 1 = parallel; 2 = not covered.

^{||}Remodeling; categories: 0 = covered; 1 = parallel or not covered.

[¶]Final model.

vitreous hemorrhage after secondary brachytherapy for 2 local recurrences (Table S5, case 10), leading to enucleation. All other eyes showed normal cosmesis at last follow-up.

Treatment-Related Side Effects

Cataract. The cumulative incidence of cataract developing or progressing, never considered to be radiation related, was comparable (50% vs. 58% at 10 years after treatment with the 10-mm and 15-mm plaque, respectively).

Iris Neovascularization. Two patients demonstrated iris neovascularization after secondary irradiation for a local recurrence (Table S5, cases 1 and 16).

Glaucoma. Two patients demonstrated glaucoma after treatment with the 10-mm plaque. The first one had had radiation maculopathy and tumor-related lipid exudation, and intraocular pressure remained elevated after vitrectomy for a rhegmatogenous retinal detachment. The second one demonstrated open-angle glaucoma. The cumulative incidence of glaucoma developing was 1% (95% CI, 0.1%–6%) at 5 and 10 years after treatment.

Vitreous Hemorrhage. Vitreous hemorrhage developed in 4 eyes a median of 1.9 years (range, 0.1–12 years) after treatment with the 10-mm plaque and in 9 eyes a median of 2.1 years (range, 0.1–13 years) after treatment with the 15-mm plaque. The cumulative incidence rates of 4% (95% CI, 1%–10%) at 5 and 10 years and 6% (95% CI, 2%–12%) at 5 years and 12% (95% CI, 5%–21%) at 10 years, respectively, were comparable ($P = 0.25$).

Maculopathy. After treatment with the 10-mm and 15-mm plaque, 17 and 20 eyes demonstrated nonproliferative radiation maculopathy, and 14 and 13 eyes demonstrated other types of maculopathy, respectively (Table 10). The cumulative incidence of both radiation maculopathy (19% [95% CI, 11%–28%]) and 18% [95% CI, 11%–27%] at 5 years and 24% [95% CI, 15%–35%] and 22% [95% CI, 14%–32%] at 10 years, respectively) and other maculopathy (4% [95% CI, 1%–11%] and 1% [95% CI, 0.1%–6%] at 5 years and 10% [95% CI, 4%–21%] and 10% [95% CI, 4%–19%] at 10 years) were comparable (Fig 3G).

Optic Neuropathy. One eye treated with the 10-mm plaque for a tumor located 4 mm from the optic disc demonstrated radiation optic neuropathy 3 years and 8 months after irradiation, whereas 7 eyes treated with the 15-mm plaque demonstrated this side effect a median of 3.1 years (range, 2.0–5.3 years) after irradiation of a tumor when its margin was a median of 2 mm (range, 0.3–2.5 mm) from the disc. The cumulative incidence of optic neuropathy developing was 1% (95% CI, 0.1%–7%) and 8% (95% CI, 3%–15%) at 5 and 10 years, respectively (Fig 3H).

Discussion

The more fundamental of the 2 questions that our study set to answer was whether treating a choroidal melanoma 10 mm or less in LBD with the 10-mm rather than with the more commonly used 15-mm ruthenium plaque is safe, using local tumor recurrence rate as the primary end point. The unadjusted rate by 5 and 10 years was 4% and 2% higher, respectively, with the 15-mm plaque, a nonsignificant difference by cumulative incidence analysis. After 10 years, 3 diffuse recurrences occurred in eyes treated with the 15-mm plaque, whereas none of the 29 melanomas treated with the 10-mm plaque still under follow-up at 10 years showed recurrence later. This might have reflected either the larger median thickness and LBD of the tumors treated with the 15-mm plaque or the higher median dose to tumor apex with the 10-mm plaque, resulting from our rule to deliver a minimum scleral dose of 250 Gy,⁶ more often realized with the comparatively thinner tumors irradiated with the 10-mm plaque. However, neither tumor size nor dose to apex was associated with local recurrence in multivariate modeling. Moreover, a sensitivity analysis that restricted tumor size to less than 9 mm to equalize size distribution returned a similar HR for plaque size as the main analysis.

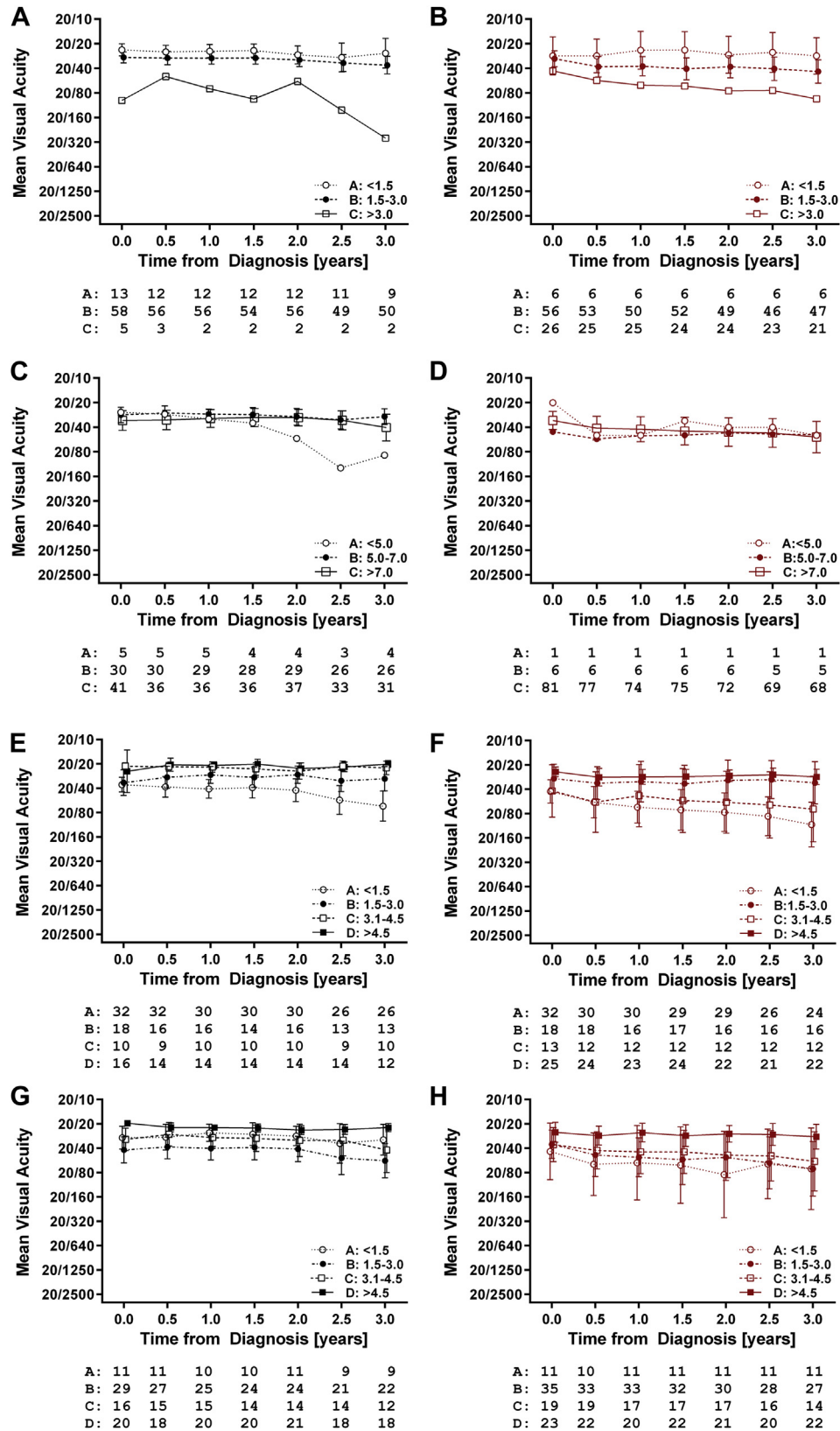


Figure 5. Graphs showing mean visual acuity for 76 and 88 patients with a choroidal melanoma less than 10 mm in largest basal diameter irradiated with the 10-mm (black) and 15-mm (red) ruthenium plaque, respectively, stratified by (A, B) tumor thickness, (C, D) largest basal tumor diameter, (E, F) distance of tumor margin to foveola, and (G, H) optic disc. Crossbars show 95% confidence intervals, omitted for groups with fewer than 7 patients.

Table 9. Cox Proportional Hazards Regression of Time to Lost Vision of 20/65 or Worse (Low Vision) and 20/400 (Blindness) in the Tumor Eye of 164 Patients with a Small Choroidal Melanoma of Whom 76 Underwent Brachytherapy with 10-mm and 88 with 15-mm Ruthenium Plaques

Variable	Low Vision				Blindness			
	Coefficient (Standard Error)	Wald Chi-Square Test	P Value	Hazard Ratio (95% CI)	Coefficient (Standard Error)	Wald Chi-Square Test	P Value	Hazard Ratio (95% CI)
Univariate analysis								
Plaque*	0.337 (0.308)	1.20	0.27	1.40 (0.77–2.56)	−0.007 (0.437)	0.98	0.99	0.99 (0.42–2.34)
Thickness†	0.266 (0.151)	3.12	0.077	1.31 (0.97–1.75)	0.422 (0.189)	4.99	0.025	1.53 (1.05–2.21)
LBD‡	−0.0169 (0.110)	0.02	0.88	0.98 (0.79–1.22)	−0.029 (0.157)	0.03	0.86	0.97 (0.71–1.32)
Dfov†	−0.428 (0.093)	21.01	<0.001	0.65 (0.54–0.78)	−0.380 (0.131)	8.36	0.004	0.68 (0.53–0.88)
BCVA baseline‡	−1.836 (0.536)	11.74	0.001	0.16 (0.06–0.46)	−1.207 (0.595)	4.11	0.043	0.30 (0.09–0.96)
Positioning§	0.038 (0.318)	0.01	0.91	1.04 (0.56–1.94)	0.315 (0.470)	0.45	0.50	1.37 (0.55–3.44)
Bivariate analysis								
Model 1		−2 log likelihood = −401.61				−2 log likelihood = −190.43		
Plaque*	0.166 (0.336)	0.24	0.62	1.18 (0.61–2.28)	−0.399 (0.488)	0.67	0.41	0.67 (0.26–1.75)
Thickness†	0.233 (0.16)	1.97	0.16	1.26 (0.91–1.75)	0.494 (0.209)	5.58	0.018	1.64 (1.09–2.47)
Model 2		−2 log likelihood = −402.33				−2 log likelihood = −195.35		
Plaque*	0.496 (0.355)	1.95	0.16	1.64 (0.82–3.30)	0.052 (0.509)	0.01	0.92	1.05 (0.39–2.86)
LBD‡	−0.108 (0.123)	0.77	0.38	0.90 (0.70–1.14)	−0.038 (0.183)	0.04	0.83	0.96 (0.67–1.38)
Model 3		−2 log likelihood = −367.46¶				−2 log likelihood = −182.58		
Plaque*	0.582 (0.310)	3.54	0.060	1.79 (0.98–3.29)	0.188 (0.439)	0.18	0.67	1.21 (0.51–2.85)
Dfov†	−0.448 (0.095)	22.08	<0.001	0.64 (0.53–0.77)	−0.385 (0.132)	8.50	0.004	0.68 (0.53–0.88)
Model 4		−2 log likelihood = −394.74				−2 log likelihood = −192.37		
Plaque*	0.330 (0.308)	1.15	0.28	1.39 (0.76–2.54)	−0.021 (0.438)	0.01	0.96	0.98 (0.41–2.31)
BCVA baseline‡	−1.864 (0.548)	11.55	0.001	0.16 (0.05–0.45)	−1.208 (0.596)	4.11	0.043	0.30 (0.09–0.96)
Trivariate analysis								
Model 5		−2 log likelihood = −366.00				−2 log likelihood = −182.0		
Plaque*	0.551 (0.310)	3.15	0.076	1.73 (0.94–3.19)	0.180 (0.439)	0.17	0.68	1.20 (0.51–2.83)
Dfov†	−0.416 (0.098)	18.10	<0.001	0.66 (0.54–0.80)	−0.361 (0.135)	7.13	0.008	0.70 (0.53–0.91)
BCVA baseline‡	−0.943 (0.753)	1.57	0.211	0.39 (0.09–1.71)	−0.587 (0.710)	0.68	0.41	0.56 (0.14–2.23)
Model 6		−2 log likelihood = −364.66				−2 log likelihood = −176.00¶		
Plaque*	0.360 (0.341)	1.13	0.29	1.44 (0.74–2.80)	−0.190 (0.474)	0.16	0.69	0.83 (0.33–2.09)
Dfov†	−0.458 (0.096)	22.64	<0.001	0.63 (0.52–0.76)	−0.438 (0.143)	9.35	0.002	0.65 (0.49–0.85)
Thickness‡	0.316 (0.186)	2.89	0.089	1.37 (0.95–1.98)	0.619 (0.232)	7.14	0.008	1.86 (1.18–2.92)

BCVA = best corrected visual acuity; CI = confidence interval; Dfov = distance to foveola; LBD = largest basal diameter.

*Categories: 0 = 10-mm plaque; 1 = 15-mm plaque.

†Continuous variable (mm).

‡Continuous variable (0.1 = logarithm of the minimum angle of resolution units).

§Remodeling; categories: 0 = covered; 1 = parallel or not covered.

¶Final model for development of low vision.

¶Final model for development of blindness.

Table 10. Type of Maculopathy for 164 Eyes with Small Choroidal Melanomas Less than 10 mm in Largest Basal Diameter Irradiated with the 10-mm and 15-mm Ruthenium Plaques

Plaque	Recurrent Tumor	Radiation Maculopathy	Radiation Scar	Epiretinal Membrane	Tumor-Related Lipid Exudation	Exudative Retinal Detachment
10 mm (n = 76)	4*	17	3	6	1	0
15 mm (n = 88)	5†	20	3	3	1	1

$P = 0.87$, Fisher exact test.

*Two primary tumors were subfoveal at presentation.

†One primary tumor was subfoveal at presentation.

The variable that expectedly predicted more frequent recurrence, adjusted to plaque size, was plaque positioning in both the main and sensitivity analysis. Both plaques were planned in approximately 40% of eyes to be parallel with the posterior tumor margin to reduce dose to the foveola. Moreover, more than 30% of the tumors were located within 2.25 mm of the margin of the optic disc, and because of the 1.5-mm thick optic nerve sheath and the nonirradiating 0.75-mm rim of the plaque, the plaque could not physically cover the entire tumor. In the latter eyes, the round plaques were chosen to limit radiation to the optic disc as compared with notched plaques. These approaches,⁶ which naturally necessitate close follow-up and immediate retreatment should evidence of a local recurrence emerge, also explain the 15% overall 10-year recurrence rate as compared with our usual rate of 10% or smaller. The 15-mm plaque covered the tumor as often as the 10-mm plaque, and multivariate analysis also showed that plaque positioning did not completely explain why the 15-mm plaque was associated with more recurrences than the 10-mm plaque. In summary, we can confidently conclude that use of the 10-mm plaque, which was preferential in our practice, most likely did not increase the rate of local recurrences.

Only a few previous studies have specifically addressed brachytherapy of choroidal melanomas in this size range. After brachytherapy with palladium, a γ -ray emitter, complete local tumor control was reported for tumors less than 10 mm in LBD mainly with round Collaborative Ocular Melanoma Study-type plaques of unspecified size.¹⁷ In contrast to our series, all tumors were 1.5 to 2.4 mm in thickness. We treated 37 and 23 tumors of this size with the 10-mm and 15-mm plaques, and 1 and 3 of them demonstrated a local recurrence, respectively. A further difference was that 39 of the 72 tumors (54%) treated with palladium were located within 6.0 mm of the optic disc and foveola, as compared with 34 of the 37 tumors (92%) and 22 of the 23 tumors (96%) treated with the 10-mm and 15-mm ruthenium plaques, respectively.

Compared with our previous report of 45 patients⁶ treated with the 10-mm plaque, rates of metastasis were, as expected, comparable with both plaques and with 31 patients more and a longer follow-up time; the 10-year rate remained 4%. The melanoma-related mortality was approximately 8% and not higher than the 6% to 10% estimate for small (T1a, stage I) choroidal melanomas in general,^{18,19} even with the 15-mm plaque, even though most tumors irradiated with it were T2a, stage IIA.

The second question that we set to answer was whether treating a choroidal melanoma of 10 mm or less in LBD with the 10-mm rather than the 15-mm ruthenium plaque conserves more vision, using as primary end points loss of 20/65 and 20/400 BCVA that correspond to development of low vision and blindness by World Health Organization criteria, respectively. Unadjusted rates of losing 20/400 vision as evaluated by cumulative incidence analysis seemed to be equal for both plaques, whereas 20/65 vision seemed to be conserved more frequently with the 10-mm plaque for the first 10 years, after which this difference vanished. Multivariate analysis confirmed that plaque size was not significant when predicting blindness, which was determined by distance of the tumor margin to the foveola and tumor thickness, controlling for plaque size.

With regard to low vision, the initial difference could have reflected the larger thickness and LBD of the melanomas treated with the 15-mm plaque. However, multivariate modeling showed that instead of tumor size, distance of its margin from the foveola in addition to plaque size was the strongest independent bivariate indicator of development of low vision, and more so than the BCVA of the treated eye at baseline. The risk of losing 20/65 vision was 1.79 times higher with the 15-mm plaque, controlling for distance to the foveola. Adding tumor thickness did not improve model fit significantly but suggested that the effect of plaque size was mediated partially by different tumor thicknesses between the 2 plaques. Again, a sensitivity analysis that restricted tumor size to less than 9 mm returned a similar HR for plaque size than the main analysis. Even if the difference in vision was marginal, the radiation scar was smaller in area with the 10-mm plaque. Our conclusion is that use of the 10-mm plaque contributed to retaining reading vision in the treated eye but could not prevent blindness in an eye with a relatively higher tumor located close to the foveola, at least during the first 10 years after treatment.

Considering the mean BCVA over the first 3 years according to Collaborative Ocular Melanoma Study conventions,¹² loss of vision from baseline was notable when tumor thickness exceeded 3.0 mm, whereas increasing LBD did not lead to vision loss. Vision remained relatively stable when treating posterior tumors located more than 1.5 mm away from foveola and the optic disc with either plaque.

The unadjusted rate of radiation maculopathy developing was equal with both plaques (19% and 18% at 5 years for the 10-mm and 15-mm plaques, respectively) as estimated

by cumulative incidence analysis and thus lower than in our earlier smaller series (28%).⁶ The unadjusted rate of radiation optic neuropathy developing was higher with the 15-mm than the 10-mm plaque but still only 9%. We confirm that radiation optic neuropathy is a rare side effect after ruthenium brachytherapy of choroidal melanomas of 10 mm or less in diameter with the 10-mm plaque, even when managing posterior tumors. The complete local tumor control with palladium plaques seems to have been offset by a higher rate of radiation maculopathy (43%) and optic neuropathy (21%)¹⁷ as compared with our results.

A limitation of our retrospective study is that a few patients had partially missing data at individual visits to referring hospitals during their later follow-up. Our study was not randomized, the 2 groups were not entirely matched, and our series was not powered to detect

increases in HR smaller than 2.0 to 2.4. Plaques were not chosen systematically for patients according to tumor characteristics. Multivariate analysis was used to adjust for these differences, which reduced bias but did not eliminate it entirely, and sensitivity analysis that restricted LBD to obtain a more similar distribution supported the observed effect sizes. These limitations taken into account, the outcomes were similar to our earlier pilot report of treatment with the 10-mm plaque.⁶ The results taken together favor continued use of the 10-mm plaque for small choroidal melanomas less than 10 mm in LBD. The local tumor recurrence rate was not worse than after the 15-mm plaque, with the suggestion of a higher chance of retaining 20/65 vision when a small choroidal melanoma of less than 10 mm in LBD is located close to the macula and optic disc.

Footnotes and Disclosures

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HUMAN SUBJECTS: Human subjects were included in this study. The study followed the tenets of the Declaration of Helsinki and was approved by the institutional review board of the Hospital Region of Helsinki and

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Author Contributions:

Conception and design: Jouhi, Kivelä

Analysis and interpretation: Jouhi, Kivelä

Data collection: Jouhi, Heikkonen, Reijonen, Raivio, Täll, Kivelä

Obtained funding: Reijonen, Kivelä; Study was performed as part of regular employment duties at the University of Helsinki and the Helsinki University Hospital. No additional funding was provided.

Overall responsibility: Jouhi, Kivelä

Abbreviations and Acronyms:

BCVA = best-corrected visual acuity; **CI** = cumulative incidence; **HR** = hazard ratio; **IOP** = intraocular pressure; **LBD** = largest basal diameter; **SE** = standard error.

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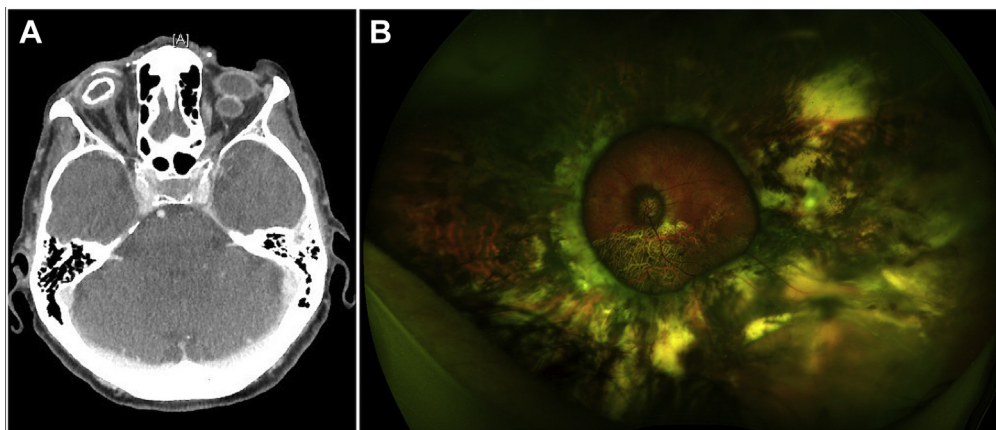
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Pictures & Perspectives



Major Scleral Buckle

A 76-year-old patient experienced a car accident several years earlier that resulted in the loss of his right eye (ocular prosthesis) and a retinal detachment in the left eye, treated with a scleral buckle. A computed tomography scan of his head (Fig A) showed the bilobed left globe separated by a ring hyperdensity (scleral buckle). Ultra-widefield fundus imaging (Fig B) demonstrated the major circular indentation delimiting the anterior and posterior retina. Significant optic disc excavation and myopic maculopathy (stage 4) were noted. Visual acuity was limited to 20/100 with narrowing of the visual field of functional and mechanical origin (Magnified version of Fig A-B is available online at www.aaojournal.org).

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